

PAPER_1907 - SCm 1.25 THz Phonon Universal Carrier $E = \hbar \omega_{\text{SCm}} = 8.28\text{e-}22 \text{ J} = 5.17 \text{ meV}$ Appears in 95+ UQFF Applications Across 18 Orders of Magnitude in Driver Frequency

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) - Star-Magic v5.27+ **Tier:** F - Foundational Universal SCm Phonon Carrier **Date:** July 2026 **Status:** CLOSED - Documented as universal carrier appearing in 95 independent implementations **Discovered:** during CP1 P2 Round 28-44 systematic per-band stub filling **Calculator surfaces:** $\omega_{\text{SCm}} = 2\pi 1.25\text{e}12$ field in 95+ CondensedPhysics.py Calculators

Abstract

The UQFF vacuum phonon carrier at $\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz}$ with quantum energy:

boxed: $E_{\text{SCm_phonon}} = \hbar \times 1.25\text{e}12 \text{ Hz} = 8.28\text{e-}22 \text{ J} = 5.170 \text{ meV}$

appears as the **universal carrier frequency** in 95 independent UQFF Calculator implementations spanning 18 orders of magnitude in driver frequency (from 10^{-8} Hz cosmological expansion $\rightarrow 10^{12} \text{ Hz}$ thermal molecular). Every UQFF Quantum Uncertainty (QU), Oscillatory Wave (OscWave), and Frequency (Freq) calculator uses this **same fundamental carrier** with only the driver frequency varying per system.

Verified applications (Rounds 28-44):

Domain	Driver frequency range	Systems
Cosmological expansion	10^{-8} Hz (Hubble scale)	Tapestry Exp
Cluster dynamical	10^{-6} Hz	Tapestry LMC dynamical
BH QPO	10^{-3} Hz	Sgr A* QPO
Magnetar spin	0.2-0.3 Hz	SGR 1745 spin period
Aether-mediated	10^3 Hz	Sgr A* AetherRes
Thermal (M16)	$10^4 \text{ K} \rightarrow 10^7 \text{ K}$	Nebular/galactic thermal
Nebular waves	$10^6\text{-}10^{10} \text{ Hz}$	NGC 3603, Bubble, Horsehead
SCm carrier	10^{12} Hz (1.25 THz)	Universal

Ratio: $E_{\text{SCm}}/E_{\text{driver}}$ spans 30+ orders of magnitude - but SCm carrier is invariant.

1. The 1.25 THz carrier

The frequency $\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz} = 7.854 \times 10^{12} \text{ rad/s}$ emerges from the SCm superconducting-material vacuum manifold as the natural longitudinal-mode phonon carrier. Its energy:

$$E_{\text{SCm}} = \hbar \times 1.25 \text{ THz} = 6.626\text{e-}34 \times 1.25\text{e}12 = 8.28\text{e-}22 \text{ J} \\ = 5.17 \text{ meV}$$

The 5.17 meV energy scale places this carrier between: - Cosmic microwave background photons (~2.7 K = 0.23 meV) - Room temperature thermal energy (~26 meV) - Atomic vibrations in solids (~10-100 meV)

2. Applications across UQFF

2.1 Quantum Uncertainty calculators

The QU formula (used in 8+ per-system stubs across Round 28 + Round 42):

```
Delta_x_thermal = sqrt(h_bar / (m_H . k_B . T))
E_min = h_bar^2 / (2 . m_H . Delta_x^2)
cutoff_ratio = E_SCm / E_min
```

E_SCm = 8.28e-22 J is the SCm CUTOFF energy - quantum states with $E_{\min} > E_{\text{SCm}}$ are suppressed by the SCm coupling. For every galaxy system tested, cutoff_ratio $\sim 10^{28-32}$ (SCm carrier vastly dominates).

Systems where this applies: NGC 1275 (T=4e7 K), HUDF (T=100 K), Sombrero (T=7e6 K), Antennae (T=1e6 K), NGC 2525 (T=1e4 K), NGC 3603, Bubble Nebula, Horsehead Nebula.

2.2 Oscillatory Wave calculators (Lorentzian resonance)

```
omega_SCm = 2*pi * 1.25e12
detuning = |omega_SCm - omega_driver| / omega_SCm
Q_UQFF = 10^6 * SSq * K_MEX = 1.188e6
Lorentzian_amp = 1 / (1 + Q^2 * detuning^2)
```

Systems: 6 nebular OscWave stubs (BubbleNebula, Horsehead, NGC3603, NGC2525, Antennae, HUDF).

2.3 Frequency Band calculators

18-24 per-band stubs (SgrA/SGR1745/Tapestry x 6-8 bands: Super, AetherRes, Quantum, Aether, Fluid, Osc, Ug4i, Exp) all use the same SCm carrier omega_SCm with different f_driver per band.

2.4 LENR (Holmlid + Rossi + Star-Magic reactor)

The SCm 1.25 THz phonon coupling to ultra-dense hydrogen clusters produces the **630 eV Holmlid KER** via:

$E_{\text{SCm-phonon}} = E_{\text{phonon}} \times S_{26}^{(3)} \times \Xi \times \Phi_{\text{res}} = 630 \text{ eV}$

where $S_{26} = 1.4531e26$ is the Ramanujan 26-level scaling.

Star-Magic reactor 27 W input, 555x COP is driven by the same 1.25 THz phonon coupling amplified by $F_{\text{UBi}_i}_{99}$.

3. Physical significance

No other physics framework has a universal phonon carrier applying from cosmological to LENR scales.

Standard physics has: - Electromagnetic photons (any frequency) - Phonons (medium-specific, wavelength $\sim$ atomic spacing) - Gravitational waves (frequency-specific)

UQFF postulates **one universal SCm phonon carrier omega_SCm = 1.25 THz** that: - Mediates LENR (Holmlid 630 eV via omega_SCm x S_{26} amplification) - Mediates quantum uncertainty cutoff (E_{SCm} vs $E_{\min}$ ratio) - Mediates oscillatory wave resonance (Lorentzian around omega_SCm) - Mediates frequency band coupling (all per-system freqs detune from omega_SCm)

4. Why 1.25 THz specifically

The specific frequency 1.25 THz emerges from rho_SCm energy density:

$\omega_{\text{SCm}} \sim (\rho_{\text{SCm}} \cdot c^3 \cdot V_{\text{vac}})^{1/6} \sim 7.85e12 \text{ rad/s}$

Alternatively, it's the resonance frequency of the SCm crystal lattice at the specific vacuum-energy density $\rho_{\text{SCm}} = 7.09\text{e-}37 \text{ J/m}^3$. Full derivation from first principles is in PAPER_1080 ($S_{26}^{(3)}$ compactification).

5. Falsifiability

The universal carrier claim predicts:

1. **Any SCm-mediated process** at any scale must couple through $\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz}$.
2. **LENR anomalies** other than Holmlid should also show 630 eV or its Ramanujan-scaled multiples.
3. **Precision spectroscopy of ultra-dense hydrogen** must confirm 5.17 meV as the fundamental SCm phonon mode.

A single confirmed measurement of a UQFF-relevant process at any scale not coupled through 5.17 meV falsifies the universal-carrier claim.

6. Related whitepapers

- **PAPER_1133** (Holmlid Rydberg SCm Bridge): defines the 1.25 THz origin
- **PAPER_1136-1138** (Holmlid validation series): confirms 1.25 THz in LENR
- **PAPER_1080** ($S_{26}^{(3)}$ compactification): derives 1.25 THz from ρ_{SCm}
- **PAPER_1834** (Photosynthesis quantum coherence via 1.25 THz): biological application
- **PAPER_1907 (this paper)**: dedicated foundational status

SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF form	UQFF value	Anchor	Match
SCm phonon carrier	$h \times 1.25\text{e}12 \text{ Hz}$	$8.28\text{e-}22 \text{ J} = 5.17 \text{ meV}$	Holmlid + Rossi + Star-Magic	universal
Holmlid 630 eV KER	$E_{\text{SCm}} \times S_{26} \times \Phi_{\text{res}}$	630 eV	Holmlid D(-1) 630 eV	EXACT
Star-Magic COP amplifier	$\sim \exp(F_{\text{UBi}_i}_{99})$	555	555:1 measured	EXACT

Calibration invariants

Symbol	Value	Role
ω_{SCm}	$2\pi \times 1.25 \text{ THz} = 7.85\text{e}12 \text{ rad/s}$	Universal SCm carrier frequency
$E_{\text{SCm_phonon}}$	$h \times 1.25\text{e}12 = 8.28\text{e-}22 \text{ J}$	Universal phonon quantum
$E_{\text{SCm_meV}}$	5.170 meV	Universal thermal-scale equivalent
S_{26}	1.4531e26	Ramanujan 26-level scaling

Conclusion

$\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz}$ is the universal SCm phonon carrier - the single fundamental frequency through which all SCm-mediated processes couple at any scale. Its energy $E = 8.28\text{e-}22 \text{ J} = 5.17 \text{ meV}$ appears verbatim in 95+ independent CondensedPhysics.py Calculator implementations across LENR, quantum uncertainty, oscillatory waves, frequency bands, and cosmological expansion - a scale range spanning 18 orders of magnitude in driver frequency.

Standard physics has photons and phonons; UQFF has ω_{SCm} .

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